Gesture Modeling and Parametrization for Autonomous Virtual Trainers

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Background



- The system under development is a two-phase immersive virtual reality training program with interactive virtual human agents.
- The system uses realistic, parameterized gestures in various contexts.
- Phase I (modeling): Experts model needed gestures/actions via demonstration using motion capture hardware.
- Phase II (training): Captured motions are re-used to train apprentice users.

Study I

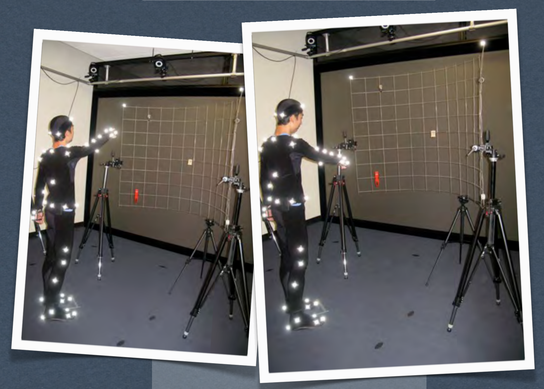
- Participants (N=25) were videotaped while pointing.
- A collection of office supplies were placed on a table.
- Manipulated: Target size, target similarity, and body position (standing/sitting).
- Observed: Hand velocity, acceleration, distance from point target.

- Singular: “Where is the [x]?
- Plural: “There is the [x] and the [y]?
- Distant points were held for longer than close points (p=.003).
- During plural points, the second point tended to be faster and held for a shorter duration.

Motion Capture

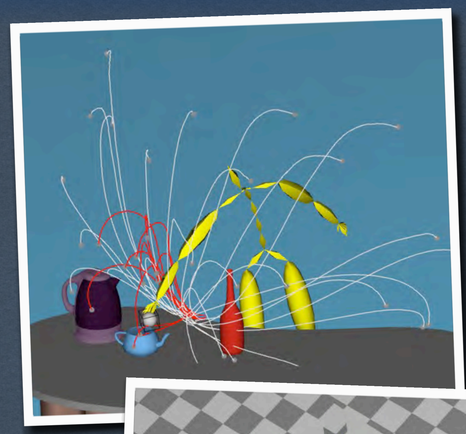


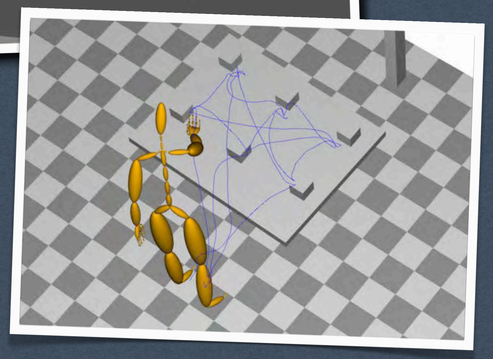
Horizontal Target Plane



Vertical Target Plane

Motion Analysis

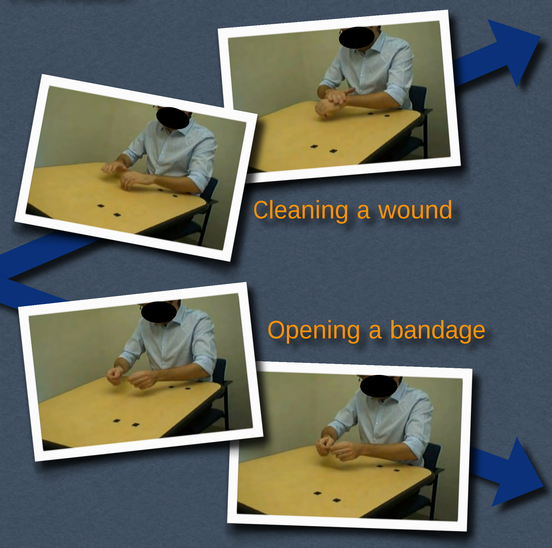




- Averaging human velocity profiles is problematic due to the high variability.
- Cluster analysis will be used to smooth the data in a structured and informative way.

Study II

- Participants were videotaped while performing both iconic and pointing gestures.
- “Please show how you would clean a wound with one hand, using a circular motion.”
  - Cleaning a wound
  - Opening a bandage
  - Applying pressure to a wound
- Substantial variability was observed across individuals.



Cleaning a wound

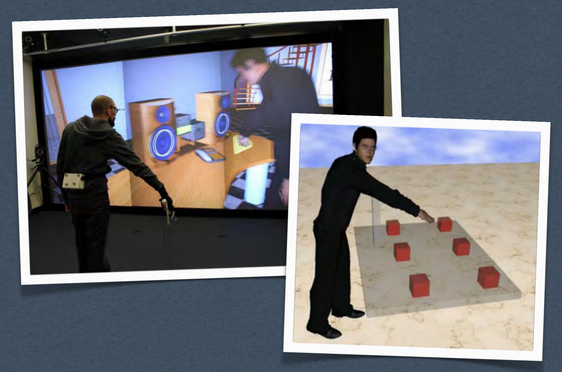
Opening a bandage

Study I – Pointing Scenario



Duct tape (L)	Mug (L)	Seashell (L)	Stapler	Scissors	Binder Clip (L)	Battery (L)
Electrical tape (S)	Mug (S)	Seashell (S)	Tape Dispenser	Tweezers	Binder Clip (S)	Battery (S)

Application



- These early findings on pointing and gesturing will inform the design and utility of interactive environments with virtual characters, and shape future work in this domain.

Funding: NSF Human-Centered Computing Grant #0915665